

AHMAD BEYH, PhD

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EMPLOYMENT

Postdoctoral Associate <i>Department of Psychiatry, Brain Health Institute, Rutgers University</i> Mentor: Dr Linden Parkes	2023–
Visiting Researcher <i>Donders Institute for Brain, Cognition and Behaviour, Radboud University</i> Mentor: Dr Stephanie Forkel	2023–2024
Postdoctoral Fellow <i>Department of Cell & Developmental Biology, University College London</i> Mentor: Prof Semir Zeki	2021–2023
Research Worker <i>Centre for Neuroimaging Sciences, King's College London</i>	2016–2021

EDUCATION

PhD, Neuroimaging <i>King's College London</i> Thesis: "The functional anatomy of white matter pathways for visual configuration learning" Supervisors: Prof Marco Catani, Prof Dominic Ffytche, Dr Flavio Dell'Acqua	2022
MSc, Neuroscience <i>King's College London</i> Thesis: "A retinotopic connectivity atlas of the human occipital lobe" Supervisor: Prof Marco Catani Graduated with distinction	2015
BA, Psychology <i>American University of Beirut</i> Thesis: "An EEG study of self-control, risk-taking, and the role of the prefrontal cortex" Supervisors: Prof Charles Harb, Prof Arne Dietrich	2014

PUBLICATIONS

* Contributed equally.

Peer-Reviewed Articles

19. Ozker M, Giglio L, Beyh A, Forkel S, Hagoort P (2025). Individual differences in speech monitoring: functional and structural correlates of delayed auditory feedback. *PNAS*.
18. Citro S, Dawson M, Beyh A, ... Catani M. Evolution of the frontal aslant tract and implications for primate vocalization and human speech. *Nature Communications*.
17. Kim JZ, Betzel R, **Beyh A**, ... Parkes L (2025). Inferring macroscopic intrinsic neural timescales using optimal control theory. *Nature Communications*.
16. Lim L, Radua J, **Beyh A** (2025). Structural connectivity abnormalities in suicidal thoughts and behaviours: a meta-analysis. *Translational Psychiatry*.

15. Mehra C, **Beyh A**, ... O'Muircheartaigh J (2025). Zero-phase-delay synchrony between interacting neural populations: implications for functional connectivity-derived biomarkers. *Imaging Neuroscience*.
14. **Beyh A**, Rasche S, Leff A, Ffytche D, Zeki S (2024). A clinico-anatomical dissection of the magnocellular and parvocellular pathways in a patient with the Riddoch syndrome. *Brain Structure & Function*.
13. **Beyh A**, Howells H, Giampiccolo D, ... Vergani F (2024). Connectivity defines the distinctive anatomy and function of the hand-knob area. *Brain Communications*.
12. Rasche S* & **Beyh A***, Paolini M, Zeki S (2024). Neural correlates of the experience of ugliness. *European Journal of Neuroscience*.
11. Zeki S, Hale ZF, **Beyh A**, Rasche S (2024). Perceptual axioms are irreconcilable with Euclidean geometry. *European Journal of Neuroscience*.
10. **Beyh A**, Rasche S, Leff A, Ffytche D, Zeki S (2023). Neural patterns of conscious visual awareness in the Riddoch syndrome. *Journal of Neurology*.
9. Rasche S* & **Beyh A***, Paolini M, Zeki S (2023). The neural determinants of abstract beauty. *European Journal of Neuroscience*.
8. O' Brien S, Sethi A, Blair J, Viding E, **Beyh A**, ... Craig MC (2023). Rapid white matter changes in children with conduct problems during a parenting intervention. *Translational Psychiatry*.
7. **Beyh A**, Dell'Acqua F, Cancemi D, De Santiago Requejo F, Ffytche D, Catani M (2022). The medial occipital longitudinal tract supports early stage encoding of visuospatial information. *Communications Biology*.
6. Mackes N, Mehta M, **Beyh A**, ... Sonuga-Barke E (2022). A prospective study of the impact of severe childhood deprivation on brain white matter in adult adoptees: widespread localized reductions in volume but unaffected microstructural organization. *eNeuro*.
5. Rajashekar D & Lavrador JP, Ghimire P, Keeble H, Harris L, Pereira N, Patel S, **Beyh A**, Gullan R, Ashkan K, Bhangoo R, Vergani F (2022). Simultaneous motor and visual intraoperative neuromonitoring in asleep parietal lobe surgery: dual-strip technique. *Journal of Personalized Medicine*.
4. Brogna C, Lavrador JP, Kandeel HS, **Beyh A**, Ribas EC, Vergani F, Tolia CM (2021). Medial-tonsillar telovelar approach for resection of a superior medullary velum cerebral cavernous malformation: anatomical and tractography study of the surgical approach and functional implications. *Acta Neurochirurgica*.
3. Mirchandani AS, **Beyh A**, Lavrador JP, Howells H, Dell'Acqua F, Vergani F (2021). Altered corticospinal microstructure and motor cortex excitability in gliomas: an advanced tractography and transcranial magnetic stimulation study. *Journal of Neurosurgery*.
2. Howells H, Thiebaut de Schotten M, Dell'Acqua F, **Beyh A**, Zappalà G, Leslie A, Simmons A, Murphy DG, Catani M (2018). Frontoparietal tracts linked to lateralized hand preference and manual specialization. *Cerebral Cortex*.
1. Catani M, Robertsson N, **Beyh A**, ... Dell'Acqua F (2017). Short parietal lobe connections of the human and monkey brain. *Cortex*.

Published Book Chapters

2. **Beyh A**, Ohlerth AK, Forkel SJ (2025). Harnessing advanced tractography in neurosurgical practice. In: Krieg SM & Picht T (Eds.), *Navigated Transcranial Magnetic Stimulation in Neurosurgery*.
1. Forkel SJ, Bortolami C, Dulyan L, Barrett R, **Beyh A** (2025). Dissecting white matter pathways: a neuroanatomical approach. In: Dell'Acqua F, Descoteaux M, Leemans A (Eds.), *Handbook of Diffusion MRI Tractography*.

Preprints & Articles Under Review

Dell'Acqua F* & **Beyh A***, ... Catani M (2025). MegaTrack: a framework for the anatomically accurate and time-efficient virtual dissection and analysis of large-scale tractography datasets. *BioRxiv*.

Matsulevits A, Alvez P, Atzori M, **Beyh A**, ... Thiebaut de Schotten M (2025). A global effort to benchmark predictive models and reveal mechanistic diversity in long-term stroke outcomes. *BioRxiv*.

Dulyan L, Bortolami C, Guzmán Chacón E, **Beyh A**, Thiebaut de Schotten M, Forkel S. Manifold lateralisation and variability in the language connectome at 7 T.

Works in Progress

Beyh A, Kim JZ, Larson B, Parkes L. Using anatomo-functional constraints to train biologically informed recurrent neural networks.

Kapteijns B, **Beyh A**, ... Forkel S. Specific white matter pathways of reading and mathematics in atypical learners: a transdiagnostic perspective on early learning and attention.

GRANTS

1K99MH145984, K99/R00 NIH Pathway to Independence Award 2026–2031
National Institute of Mental Health
Title: Computational modeling of the brain structure-function links underlying hallucinations in psychosis and Parkinson's disease
Role: PI
Under review

FELLOWSHIPS, AWARDS, & SCHOLARSHIPS

Fellowships

Fellow, The Philadelphia Symposium for Postdoctoral Initiatives in Neuroscience Excellence 2025

Awards & Honors

Best Poster Award, Brain Health Institute, Rutgers University 2025
Best Abstract Award, European Workshop on Cognitive Neuropsychology 2018
Finalist, Wellcome Image Awards, The Wellcome Trust 2017
The Henry Mc Ilwain Best Poster Award, King's College London 2015

Scholarships & Travel Support

Travel Grant, Brain Health Institute, Rutgers University 2025
Scholarship, CIFAR Neuroscience of Consciousness Winter School 2022
Travel Grant, Centre for Doctoral Studies, King's College London 2019
Travel Grant, Guarantors of Brain 2019

TEACHING & SPEAKING ENGAGEMENTS

Guest Lectures

Networks of the visual system (King's College London) 2025–2026
Introduction to neuroimaging (University College London) 2020–2022

Occipital lobe anatomy & function (The London Neuromonitoring and Brain Mapping Course)	2022
Neuroanatomy for neuroimagers (King's College London)	2020
Introduction to diffusion MRI (King's College London)	2019
<i>Attendee score: 4.7/5</i>	
Networks of the visual system (King's College London)	2016–2018

Invited Talks & Research Seminars

Bridging neuroscience & AI with biologically constrained neural networks (University of Pennsylvania)	2025
From diffusion MRI to white matter pathways (Rutgers University)	2025
<i>Attendee score: 4.7/5</i>	
bioRNNs: toward more biologically plausible neural network models of the brain (Rutgers University)	2025
Medial occipito-temporal networks (University of Bordeaux)	2023
Callosal connections and visual motion processing (Neuroccino, Clinical Neuroanatomy Seminars)	2023
Occipital lobe anatomy & function (The London Neuromonitoring and Brain Mapping Course)	2022
GPS and spatial memory (Neuroccino, Clinical Neuroanatomy Seminars)	2022

Workshops & Technical Instruction

Introduction to diffusion MRI preprocessing, NEURAL Workshop (virtual)	2025
<i>Attendee score: 5/5</i>	
Diffusion MRI, Neuroanatomy, and Tractography Workshop (University of Auckland)	2023–2024
<i>Attendee score: 4.6/5</i>	
Diffusion MRI preprocessing, Symposium on Diffusion Imaging (UCLouvain)	2018
Neuroanatomy and Tractography Workshop, NatBrainLab (King's College London)	2016–2018

ACADEMIC SERVICE & LEADERSHIP

Editorial Roles

Consulting Editor, Cortex	2025–
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Peer Review — Scientific Journals

Brain Communications · Brain Structure & Function · Communications Biology · Cortex · eLife · Human Brain Mapping · Journal of Headache and Pain · Journal of Neuroscience · Nature Communications · Network Neuroscience · NeuroImage · Neurosurgical Review · PNAS · Scientific Reports

Peer Review — Funding Agencies

Brain Research UK · French National Research Agency

Committee Roles

Secretary, Neuroscience Alliance	2024–
Organizer, Neural Networks Journal Club, Rutgers University	2023–
Member, Organizing Committee, Neuroanatomy and Tractography Workshop, NatBrainLab	2016–2021
Member, Organizing Committee, Clinical Neuroanatomy Seminars	2015–2018

Member, Organizing Committee, London BrainHack	2017
<i>Outreach & Community Engagement</i>	
Scientific Consultant, National Art Pass Campaign, Art Fund UK	2023
Panelist, Discover Careers in Clinical Practice, King's College London	2021
Panelist, IoPPN Alumni Career Event, King's College London	2015
Member, Organizing Committee, Autism Awareness Campaign, American University of Beirut	2014
CONFERENCE PRESENTATIONS	
<i>Contributed Talks</i>	
American Physical Society (Anaheim, CA, USA)	2025
CIFAR Neuroscience of Consciousness Winter School (Cancun, Mexico)	2022
International Society for Magnetic Resonance in Medicine, Diffusion Study Group (virtual)	2021
Gray's Neuroanatomy Conference (Siracusa, Sicily, Italy)	2018
European Workshop on Cognitive Neuropsychology (Bressanone, Italy)	2018
<i>Posters</i>	
Rutgers Brain Health Institute Annual Symposium (New Brunswick, NJ, USA)	2025
Organization for Human Brain Mapping (Rome, Italy)	2019
Organization for Human Brain Mapping (Vancouver, Canada)	2017
Organization for Human Brain Mapping (Geneva, Switzerland)	2016
Neuroimaging and Experimental Medicine Advances (London, UK)	2015
PROFESSIONAL DEVELOPMENT	
Deep Learning Course, Neuromatch Academy	2024
The Human Connectome Project Course	2018
Preparing to Teach in Higher Education, King's College London	2016
The FSL Course, Oxford Centre for Functional MRI of the Brain	2016